

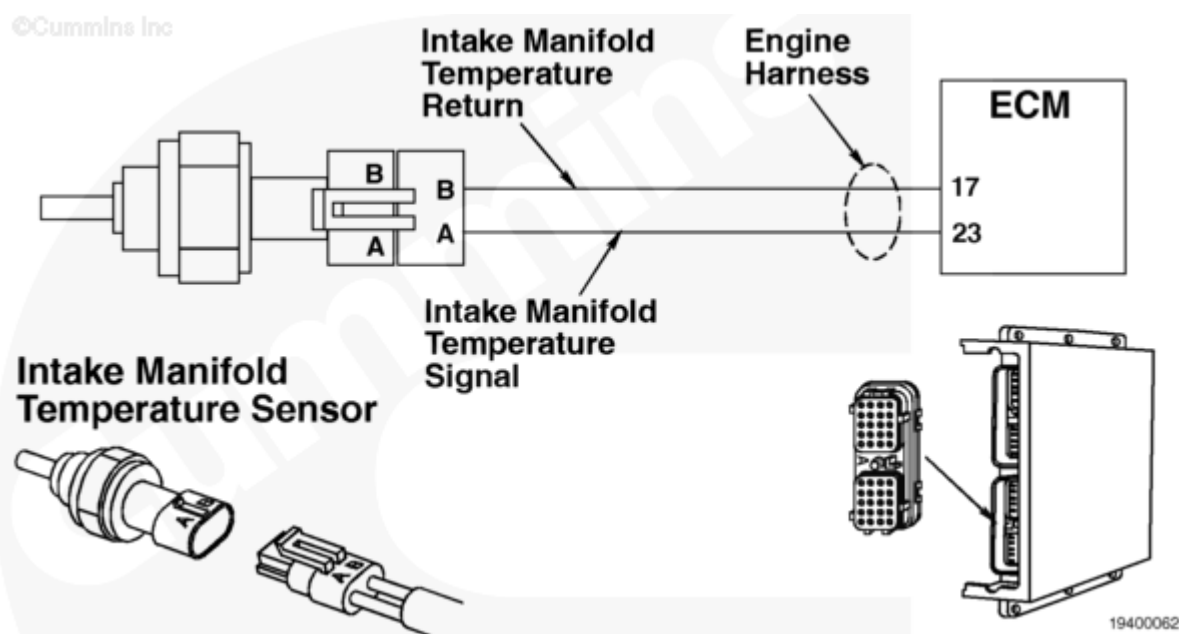
## Fault Code: 154

### Intake Manifold Temperature Sensor Circuit

 Printable Version

#### Overview

| Codes                                                                                    | Reason                                                                                                  | Effect                                                        |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Fault Code: 154<br><br>PID(P): P105<br>SPN: 105<br>FMI: 4<br>Lamp: Yellow<br>SRT: 00-359 | Less than 0.08 VDC detected at the intake manifold air temperature signal pin 23 of the engine harness. | No engine protection for the intake manifold air temperature. |



Intake Manifold Temperature Sensor Circuit

#### Circuit Description

The intake manifold temperature sensor is used by the ECM to monitor the temperature of the air in the intake manifold downstream of the aftercooler element. The intake manifold temperature sensor is used by the ECM for the engine protection system and timing and fueling control. The ECM monitors the voltage on pin 23. The ECM expects to see the voltage vary between 0.5 and

4.5 VDC. If the voltage is below 0.08 VDC, then the ECM will log Fault Code 154. Voltage below 0.08 VDC on pin 23 can be caused by shorts to ground on the supply or return wires, or an internally grounded, failed sensor.

## Component Location

QSK19 series engines - The intake manifold temperature sensor is located above the fuel pump, next to the intake manifold pressure sensor.

## Shoptalk

All temperature sensors:

- The resistance of the sensor varies with the temperature. The reading that you observe could possibly compare to the following table if the sensor is functioning properly.

| Temperature(°C) | Temperature[°F] | Resistance(ohms) |
|-----------------|-----------------|------------------|
| 0               | 32              | 30k to 36k       |
| 25              | 77              | 9k to 11k        |
| 50              | 122             | 3k to 4k         |
| 75              | 167             | 1350 to 1500     |
| 100             | 212             | 600 to 675       |

Refer to Troubleshooting Fault Code t05-154 (</qs3/pubsys2/xml/en/procedures/19/19-t05-154.html>)